

RANTOUL, IL, USA

WMO: 722194

Lat: 40.293N

Lon: 88.142W

Elev: 737

StdP: 14.31

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 04896

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -0.1	(c) 4.8	(d) -8.0	(e) 3.7	(f) 2.8	(g) -3.4	(h) 4.7	(i) 6.5	(j) 28.7	(k) 31.6	(l) 26.4	(m) 30.6	(n) 7.5	(o) 310	(p) 0.509

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.7	92.5	75.7	90.4	75.0	88.2	73.8	81.0	87.5	78.7	86.0	76.9	84.4	10.0	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.1	155.5	85.6	76.8	143.3	83.6	74.6	133.3	80.9	45.2	87.5	42.8	85.7	40.7	84.6	91.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 25.8	(b) 22.8	(c) 19.6	(e) -7.2	(f) 95.6	(g) 6.4	(h) 2.9	(i) -11.9	(j) 97.6	(k) -15.6	(l) 99.3	(m) -19.3	(n) 100.9	(o) -23.9	(p) 103.0		
(4) 25.8	(5) 22.8	(6) 19.6	(7) -7.2	(8) 93.7	(9) 6.7	(10) 3.1	(11) -12.0	(12) 86.0	(13) -15.9	(14) 87.8	(15) -19.7	(16) 89.5	(17) -24.6	(18) 91.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	53.2	26.3	29.0	41.6	53.4	64.9	73.5	75.2	73.8	68.5	55.7	42.7	32.0
	DBStd	19.82	12.09	11.97	11.33	9.34	9.16	5.83	5.41	5.29	7.06	9.45	9.55	10.67
	HDD50	2583	736	594	314	69	6	0	0	0	0	45	255	564
	HDD65	5490	1199	1009	730	362	121	6	1	5	43	320	670	1024
	CDD50	3740	2	6	52	170	467	704	782	739	556	221	36	5
	CDD65	1170	0	0	3	13	117	261	318	278	148	31	1	0
	CDH74	11031	0	0	32	189	1187	2438	3015	2458	1402	307	3	0
	CDH80	3917	0	0	3	28	419	879	1143	847	515	83	0	0
Wind	WSAvg	8.6	10.7	10.3	10.4	10.8	9.0	7.4	5.6	5.3	6.3	8.4	9.7	9.9
Precipitation	PrecAvg	39.4	2.4	2.0	2.7	3.7	4.4	4.3	4.3	4.1	2.6	3.4	3.3	2.4
	PrecMax	55.6	6.6	5.0	6.2	8.9	8.4	9.1	9.6	7.1	7.6	7.0	7.6	6.3
	PrecMin	31.5	0.6	0.3	0.7	1.5	1.3	1.0	1.0	2.2	0.6	1.3	0.6	1.1
	PrecStd	6.0	1.4	1.4	1.5	1.7	1.9	1.9	2.2	1.7	1.7	1.7	1.6	1.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	59.7	65.2	78.8	83.1	91.7	94.4	97.7	94.9	94.7	87.9	72.2	62.9
		MCWB	57.0	58.3	63.6	61.7	69.9	75.0	76.5	76.7	71.4	69.5	60.8	58.8
	2%	DB	53.8	58.4	70.4	79.5	88.5	90.9	91.5	90.8	90.3	81.7	65.9	56.6
		MCWB	51.1	54.3	59.8	60.5	68.5	75.1	78.1	76.4	72.0	64.6	58.6	53.6
	5%	DB	47.8	50.5	64.0	75.1	84.7	88.5	89.9	88.2	86.5	77.0	61.4	50.9
		MCWB	45.6	46.7	56.5	60.2	66.9	73.4	78.3	74.9	71.9	63.0	55.3	48.4
	10%	DB	42.1	45.1	59.4	70.2	81.2	85.6	86.4	84.7	81.9	71.9	57.2	46.2
		MCWB	39.5	41.7	53.0	58.5	66.1	71.7	75.2	73.1	67.1	61.8	51.1	43.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	57.5	60.4	64.5	68.4	75.6	82.1	85.3	83.4	78.3	73.4	64.4	60.0
		MCDB	60.0	63.8	74.4	74.9	85.4	89.8	89.7	89.4	87.7	82.5	69.7	62.0
	2%	WB	51.5	53.9	61.4	65.3	73.0	79.3	81.7	79.8	76.0	69.7	60.2	54.0
		MCDB	53.0	57.8	69.8	73.9	82.2	87.2	87.9	86.3	84.9	77.0	64.0	56.2
	5%	WB	45.6	47.7	58.3	62.6	71.0	77.1	79.1	77.5	73.8	66.9	56.3	48.5
		MCDB	47.5	50.3	63.6	70.5	79.8	84.5	86.0	84.8	81.2	73.2	61.0	50.4
	10%	WB	39.5	41.4	53.9	59.9	68.9	74.9	77.2	75.4	71.5	63.5	52.2	44.1
		MCDB	41.9	45.3	58.2	68.2	77.1	81.7	84.6	82.2	78.4	70.9	56.5	46.1
Mean Daily Temperature Range	5% DB	MDBR	14.6	15.6	17.9	21.1	21.0	19.4	18.7	19.5	21.7	20.5	17.1	13.2
		MCDBR	20.0	22.2	25.1	27.7	25.6	22.7	21.0	22.2	26.1	26.8	23.7	18.6
		MCWBR	17.9	18.1	16.3	13.9	10.8	10.6	10.7	11.2	10.9	13.1	16.4	16.3
	5% WB	MCDBR	19.2	21.7	22.5	22.6	21.9	19.1	18.7	19.3	21.2	21.1	21.3	17.6
MCWBR		18.1	18.9	15.7	14.9	11.8	11.2	11.9	11.7	10.9	13.0	16.9	16.2	
Clear-Sky Solar Irradiance	taub	0.300	0.315	0.352	0.394	0.435	0.441	0.458	0.455	0.415	0.362	0.315	0.297	
	taud	2.432	2.383	2.365	2.252	2.194	2.221	2.155	2.152	2.243	2.388	2.476	2.453	
	Ebn at Noon	274	285	285	279	269	266	260	258	260	265	266	266	
	Edh at Noon	25	31	35	42	46	44	47	46	39	30	24	23	
All-Sky Solar Radiation	RadAvg	564	846	1157	1503	1718	1969	1951	1731	1459	995	662	474	
	RadStd	63	82	90	114	145	123	102	105	109	105	62	58	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (51 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page